

Computing Theory and Practices (2026)

Course code 24PY902

M1U1

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M1U1 Plan

- 1 Introduction
- 2 Linear Algebra Review
- 3 Vectors and vector spaces
- 4 Matrices and matrix transformations

Learning Outcomes

After this lecture students will be able to

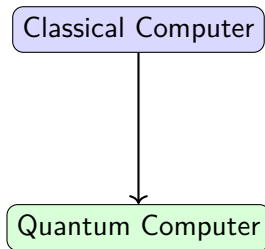
- Understand the motivation for quantum computing.
- Explain limitations of classical computing.
- Describe the evolution of quantum computation.
- Identify major application areas.
- Appreciate current quantum hardware platforms.
- Understand the structure of this course.

Why Quantum Computing?

Classical computers have transformed society, yet several important problems remain computationally expensive.

Examples include

- Cryptography
- Drug discovery
- Materials design
- Optimization
- Machine learning
- Molecular simulation



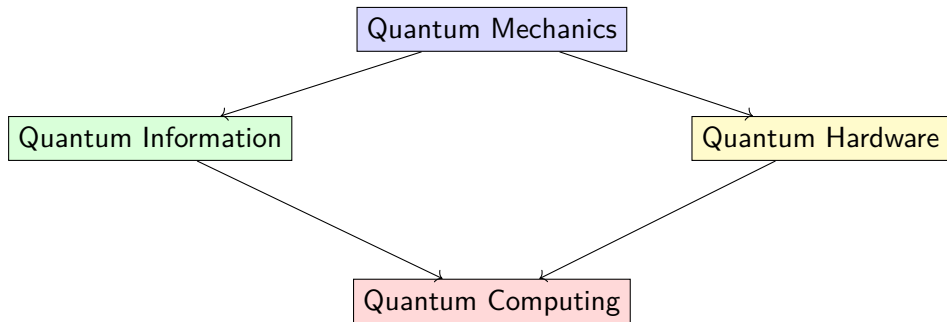
Classical vs Quantum Computing

Feature	Classical	Quantum
Information	Bit	Qubit
State	0 or 1	Superposition
Correlation	Classical	Entanglement
Operations	Logic Gates	Quantum Gates
Parallelism	Limited	Quantum Parallelism
Applications	General Purpose	Specialized

Historical Milestones

- 1900 – Planck introduces quantum theory
- 1925 – Heisenberg develops Matrix Mechanics
- 1926 – Schrödinger wave mechanics
- 1981 – Richard Feynman proposes quantum simulation
- 1985 – David Deutsch formulates universal quantum computer
- 1994 – Peter Shor introduces factoring algorithm
- 1996 – Lov Grover develops quantum search
- Today – IBM, Google, IonQ, Rigetti, Quantinuum and others build quantum processors

Quantum Computing Ecosystem



Major Application Areas

- Quantum Chemistry
- Drug Discovery
- Financial Modeling
- Logistics
- Artificial Intelligence
- Cybersecurity
- Climate Modeling
- Optimization
- Secure Communication
- Material Science
- Quantum Internet
- National Security

Course Roadmap

- 1 Linear Algebra
- 2 Quantum States
- 3 Operators
- 4 Hilbert Space
- 5 Quantum Hardware
- 6 Qubits
- 7 Quantum Gates
- 8 Quantum Algorithms
- 9 Quantum Circuits
- 10 Practical Qiskit Programming

Summary

- Classical computing has practical limitations.
- Quantum mechanics enables a new computational paradigm.
- Qubits exploit superposition and entanglement.
- Quantum computing has applications across science and engineering.
- This course develops both theoretical foundations and practical implementation.

Practice Questions

Discussion

- 1 Why do classical computers struggle with molecular simulation?
- 2 What distinguishes a qubit from a classical bit?
- 3 Name three applications of quantum computing.
- 4 Who proposed the universal quantum computer?
- 5 Why is quantum computing expected to complement rather than replace classical computing?

Reading

- Nielsen & Chuang — Chapter 1
- IBM Quantum Learning

Learning Outcomes

After this lecture, students will be able to

- Recall the fundamentals of vectors and matrices.
- Understand vector spaces and basis vectors.
- Compute dot products and vector norms.
- Perform matrix operations.
- Appreciate why linear algebra forms the language of quantum mechanics.

Why Linear Algebra?

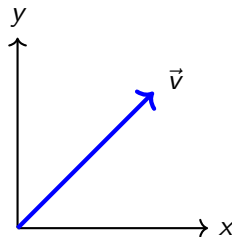
Quantum mechanics represents every physical state as a vector in a vector space.

Every quantum computation consists of

- State vectors
- Operators
- Matrices
- Eigenvalues
- Inner products
- Basis transformations

Therefore,

Quantum Mechanics $\implies$ Linear Algebra



Vectors and Vector Spaces

A vector is an ordered collection of numbers.

Examples

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Vector addition

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

Scalar multiplication

$$3\mathbf{v} = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

Dot Product and Vector Norm

For two vectors

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Dot Product

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2$$

Vector Length

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}$$

Example

$$\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

then

Matrices and Matrix Operations

A matrix is a rectangular arrangement of numbers.

Example

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 0 & 5 \end{bmatrix}$$

Matrix Addition

$$A + B = \begin{bmatrix} 3 & 3 \\ 3 & 9 \end{bmatrix}$$

Scalar Multiplication

$$2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Remark: Quantum gates are represented by matrices acting on state vectors.

Matrix Multiplication

Given

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$$

Matrix multiplication is performed by multiplying rows of the first matrix with columns of the second matrix.

$$AB = \begin{bmatrix} 1(2) + 2(1) & 1(0) + 2(5) \\ 3(2) + 4(1) & 3(0) + 4(5) \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}$$

Important Property

$$AB \neq BA$$

In general, matrix multiplication is *not commutative*.

Eigenvalues and Eigenvectors

For a square matrix A ,

$$A\mathbf{x} = \lambda\mathbf{x}$$

where

- $\mathbf{x}$ is an eigenvector.
- λ is the corresponding eigenvalue.

Example

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Then

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Hence,

[1]

Linear Algebra in Quantum Computing

Example of a qubit	
Linear Algebra	Quantum Computing
Vector	Quantum State
Matrix	Quantum Gate
Basis	Computational Basis
Inner Product	Probability Amplitude
Eigenvalue	Measurement Outcome
Identity Matrix	Identity Gate

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

subject to

$$|\alpha|^2 + |\beta|^2 = 1$$

Every quantum algorithm manipulates vectors using matrices.

Summary and Practice

Key Takeaways

- Linear algebra is the mathematical language of quantum mechanics.
- Quantum states are vectors.
- Quantum gates are matrices.
- Matrix multiplication represents quantum evolution.
- Eigenvalues and eigenvectors describe quantum measurements.

Practice Questions

- 1 Add the vectors $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$.
- 2 Find the norm of $\begin{bmatrix} 6 \\ 8 \end{bmatrix}$.
- 3 Compute the product of two 2×2 matrices.
- 4 Explain why $AB \neq BA$ in general.
- 5 Why are eigenvalues important in quantum mechanics?

Recommended Reading

- Nielsen & Chuang, *Quantum Computation and Quantum Information*, 

Learning Outcomes

After this lecture, students will be able to

- Explain the concept of vector spaces.

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- Test orthogonality of vectors.

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After this lecture, students will be able to

- Explain the concept of vector spaces.
- Distinguish vectors from scalars.
- Compute inner (dot) products.
- Test orthogonality of vectors.
- Understand basis vectors used to represent qubits.

Motivation

Why do we need vector spaces in quantum mechanics?

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Probabilities are calculated using inner products.

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Probabilities are calculated using inner products.

Therefore,

Quantum Computing = Linear Algebra + Quantum Physics

Vectors

A vector possesses both

- Magnitude

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- Magnitude
- Direction

Example

$$\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Magnitude

$$|\mathbf{v}| = \sqrt{3^2 + 4^2} = 5$$

Vectors

A vector possesses both

- Magnitude
- Direction

Example

$$\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Magnitude

$$|\mathbf{v}| = \sqrt{3^2 + 4^2} = 5$$

A normalized vector has unit magnitude.

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

Vector Spaces

A vector space is a collection of vectors satisfying

- 1 Closure under addition

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Examples

- Two-dimensional plane

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- Two-dimensional plane
- Three-dimensional space

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Examples

- Two-dimensional plane
- Three-dimensional space
- Complex vector space used in quantum mechanics

Inner Product and Geometric Interpretation

For two vectors

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

the inner (dot) product is defined as

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2$$

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$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta$$

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Interpretation

- Large positive value $\rightarrow$ vectors point in similar directions.
- Zero $\rightarrow$ vectors are perpendicular.
- Negative value $\rightarrow$ vectors point in opposite directions.

Orthogonality and Orthonormal Basis

Two vectors are said to be orthogonal if

$$\mathbf{u} \cdot \mathbf{v} = 0$$

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Example

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

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Orthogonality and Orthonormal Basis

Two vectors are said to be orthogonal if

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Example

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\mathbf{u} \cdot \mathbf{v} = 0$$

An orthonormal basis satisfies

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$$

where

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Computational Basis States

Quantum computing uses an orthonormal basis called the **computational basis**.

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$$\langle 1|1\rangle = 1$$

$$\langle 0|1\rangle = 0$$

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$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These satisfy

$$\langle 0|0\rangle = 1$$

$$\langle 1|1\rangle = 1$$

$$\langle 0|1\rangle = 0$$

Any qubit can be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

subject to

Linear Algebra Concept

- Vector

Linear Algebra Concept

- Vector
- Inner Product

Linear Algebra Concept

- Vector
- Inner Product
- Matrix

Linear Algebra Concept

- Vector
- Inner Product
- Matrix
- Basis

Linear Algebra Concept

- Vector
- Inner Product
- Matrix
- Basis
- Eigenvector

Quantum Interpretation

- Quantum State

Application to Quantum Computing

Linear Algebra Concept

- Vector
- Inner Product
- Matrix
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Quantum Interpretation

- Quantum State
- Probability Amplitude

Application to Quantum Computing

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Quantum Interpretation

- Quantum State
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Linear Algebra Concept

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- Inner Product
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Quantum Interpretation

- Quantum State
- Probability Amplitude
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- Computational Basis
- Measurement State

Application to Quantum Computing

Linear Algebra Concept

- Vector
- Inner Product
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- Basis
- Eigenvector

Quantum Interpretation

- Quantum State
- Probability Amplitude
- Quantum Gate
- Computational Basis
- Measurement State

Key Idea

Every quantum algorithm is a sequence of matrix operations acting on vectors in a Hilbert space.

Summary, Concept Check and References

Summary

- Quantum states are vectors.
- Inner products determine probabilities and orthogonality.
- Vector spaces provide the mathematical framework for quantum mechanics.
- Computational basis states are orthonormal vectors.
- Linear algebra is the foundation of quantum computation.

Concept Check

- 1 What is a vector space?
- 2 When are two vectors orthogonal?
- 3 Why is normalization important for a quantum state?
- 4 What are the computational basis states?
- 5 How do matrices represent quantum gates?

References

- Nielsen & Chuang, *Quantum Computation and Quantum Information*.
- Gilbert Strang, *Introduction to Linear Algebra*

Learning Outcomes

After this lecture students will be able to

- Describe matrices as linear operators.

Learning Outcomes

After this lecture students will be able to

- Describe matrices as linear operators.
- Perform matrix addition and multiplication.

Learning Outcomes

After this lecture students will be able to

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Learning Outcomes

After this lecture students will be able to

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- Visualize scaling, rotation and reflection.

Learning Outcomes

After this lecture students will be able to

- Describe matrices as linear operators.
- Perform matrix addition and multiplication.
- Explain linear transformations.
- Visualize scaling, rotation and reflection.
- Relate quantum gates to matrices.

Why Matrices?

Matrices provide a compact way to describe transformations.

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Examples include

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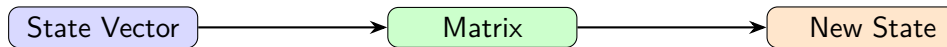
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Matrices provide a compact way to describe transformations.

Examples include

- Image rotation
- Robot motion
- Computer graphics
- Quantum gate operations



Matrix Representation

A matrix is a rectangular array of numbers.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

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A vector

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

is transformed as

$$\boxed{A\mathbf{x} = \mathbf{y}}$$

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is transformed as

$$\boxed{A\mathbf{x} = \mathbf{y}}$$

The matrix acts like a machine that transforms one vector into another.

Linear Transformations

Example: Scaling Transformation

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

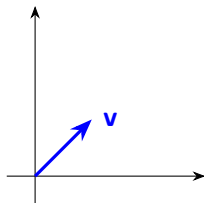
Linear Transformations

Example: Scaling Transformation

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

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$$S\mathbf{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$



Scaling changes magnitude while preserving direction.

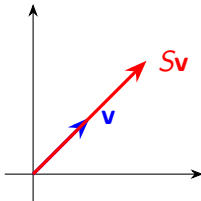
Linear Transformations

Example: Scaling Transformation

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$S\mathbf{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$



Scaling changes magnitude while preserving direction.

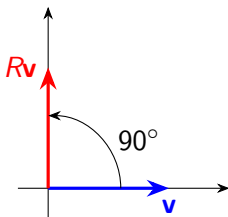
Rotation Transformation

A rotation changes the direction of a vector while preserving its magnitude. For an angle θ , the rotation matrix is

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Example:

$$R(90^\circ) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$



Rotation matrices preserve vector length and angles

Reflection Transformation

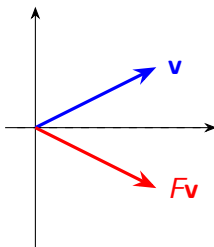
Reflection mirrors a vector about an axis.

Reflection about the x-axis is represented by

$$F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Applying the transformation,

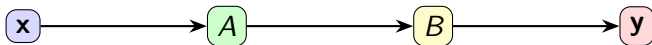
$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$



Matrix Multiplication as Composition

Successive linear transformations are represented by matrix multiplication.

$$\mathbf{y} = B(A\mathbf{x}) = (BA)\mathbf{x}$$



Important

$$AB \neq BA$$

Changing the order of transformations generally changes the result.

Quantum Gates as Linear Transformations

Quantum gates are matrices acting on quantum state vectors.

Identity Gate

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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Pauli-X Gate

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Quantum Gates as Linear Transformations

Quantum gates are matrices acting on quantum state vectors.

Identity Gate

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Pauli-X Gate

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Hadamard Gate

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Quantum Gates as Linear Transformations

Quantum gates are matrices acting on quantum state vectors.

Identity Gate

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Pauli-X Gate

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Hadamard Gate

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Example

$$X|0\rangle = |1\rangle$$

Summary and Concept Check

Summary

- Matrices represent linear transformations.
- Scaling, rotation and reflection are matrix operations.
- Matrix multiplication combines successive transformations.
- Matrix multiplication is generally non-commutative.
- Quantum gates are matrices acting on qubits.

Concept Check

- 1 What is a linear transformation?
- 2 Which transformation preserves vector magnitude?
- 3 Why is matrix multiplication not commutative?
- 4 Which quantum gate exchanges $|0\rangle$ and $|1\rangle$?
- 5 Why are matrices central to quantum computing?

Suggested Reading

- Nielsen & Chuang, *Quantum Computation and Quantum Information*.
- Gilbert Strang, *Introduction to Linear Algebra*.
- IBM Quantum Learning

Learning Outcomes

After this lecture, you will be able to

- Define a matrix and identify its order.
- Classify common types of matrices.
- Perform basic matrix operations.
- Understand matrix multiplication.
- Explain matrices as linear transformations.
- Relate matrices to quantum gates.

Review: Scalars, Vectors and Matrices

Scalar

$5, -2, 3 + i$

Single numerical value.

Vector

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Represents magnitude and direction.

Matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Represents a linear transformation.

Key Idea

Matrices transform vectors into new vectors.

What is a Matrix?

A matrix is a rectangular arrangement of numbers in rows and columns.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

This matrix has

2 rows 3 columns

Hence,

$$A_{2 \times 3}$$

Matrices are represented using capital letters.

A, B, C, M, U

Common Types of Matrices

Row Matrix $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$

Column Matrix $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

Square Matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Zero Matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Identity Matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

The identity matrix behaves like the number 1 in ordinary arithmetic.

$$I\mathbf{x} = \mathbf{x}$$

Basic Matrix Operations

For

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Addition

$$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

Scalar Multiplication

$$2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Only matrices having the same dimensions can be added.

Matrix Multiplication

For two matrices

$$A_{m \times n}, \quad B_{n \times p}$$

the product

$$AB$$

exists only when the number of columns of A equals the number of rows of B .

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Important

Matrix multiplication is generally

$$AB \neq BA$$

Therefore, matrix multiplication is ****not commutative****

Matrices as Linear Transformations

A matrix transforms one vector into another.

$$\mathbf{y} = A\mathbf{x}$$

where

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Then,

$$\mathbf{y} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

A matrix may rotate, reflect, stretch or shrink a vector.

Matrix Representation in Quantum Computing

Every quantum gate is represented by a matrix.

Examples

$$\text{Identity Gate} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{NOT (Pauli-X) Gate} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Quantum computation is performed by multiplying a gate matrix with a qubit state vector.

$$|\psi'\rangle = U|\psi\rangle$$

where U is a unitary matrix.

Worked Example, Summary and Concept Check

Worked Example

Given

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

find

$$A\mathbf{x}$$

Solution

$$A\mathbf{x} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$

Summary

- Matrix definition
- Types of matrices

Learning Outcomes

After this lecture, you will be able to

- Explain the concept of a quantum state.
- Define a Hilbert space.
- Describe computational basis states.
- Explain quantum superposition.
- Apply the normalization condition.
- Visualize a qubit using the Bloch sphere.

From Classical States to Quantum States

Classical Bit

- State = 0 or 1
- One definite value
- Deterministic

0 or 1

Quantum Bit

- Exists in superposition
- Described by a vector
- Probabilistic measurement

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Key Idea

A quantum state contains all the information about a quantum system.

What is a Quantum State?

A quantum state is the mathematical description of a quantum system.

For a single qubit,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where

- α and β are complex probability amplitudes.
- $|0\rangle$ and $|1\rangle$ are basis states.

The state vector predicts the outcome probabilities of measurements.

Hilbert Space

A Hilbert space is a complete complex vector space equipped with an inner product.

A Hilbert space satisfies

- Vector addition
- Scalar multiplication
- Inner product
- Normalization
- Orthogonality

For a single qubit,

$$\mathcal{H} = \mathbb{C}^2$$

That is, the state of one qubit lives in a two-dimensional complex vector space.

Computational Basis States

The computational basis consists of two orthonormal vectors.

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These vectors satisfy

$$\langle 0|1\rangle = 0$$

and

$$\langle 0|0\rangle = \langle 1|1\rangle = 1$$

They form the basis for representing every single-qubit state.

Quantum Superposition

Unlike a classical bit, a qubit can exist in a superposition of basis states.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Interpretation

- α determines the contribution of $|0\rangle$
- β determines the contribution of $|1\rangle$
- Measurement collapses the state into one of the basis states.

Superposition is one of the fundamental principles of quantum mechanics.

Normalization Condition

The total probability of all possible outcomes must equal one.

Therefore,

$$|\alpha|^2 + |\beta|^2 = 1$$

Example

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

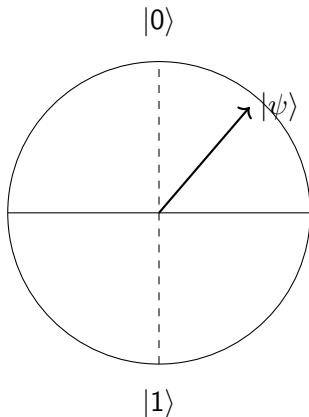
Since

$$\frac{1}{2} + \frac{1}{2} = 1$$

the state is normalized.

Bloch Sphere Representation

Every normalized single-qubit state can be represented as a point on the Bloch sphere.



The Bloch sphere provides a geometric picture of a single-qubit state.

Summary and Concept Check

Summary

- Quantum states are represented by vectors.
- Hilbert space is the mathematical framework of quantum mechanics.
- Computational basis states are $|0\rangle$ and $|1\rangle$.
- Superposition allows a qubit to exist in multiple states.
- Every quantum state must be normalized.
- The Bloch sphere visualizes single-qubit states.

Concept Check

- 1 What is a Hilbert space?
- 2 Why must quantum states be normalized?
- 3 What is the difference between a basis state and a superposition state?
- 4 Why is the Bloch sphere useful?

Learning Outcomes

After this lecture, you will be able to

- Represent quantum states using Dirac notation.
- Distinguish between bras and kets.
- Calculate inner products.
- Explain orthonormal basis states.
- Represent multi-qubit states.
- Interpret quantum states geometrically.

Review: Quantum States and Hilbert Space

A single-qubit quantum state is represented as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

A Hilbert space is

- a complex vector space,
- equipped with an inner product,
- complete under vector operations.

For one qubit,

$$\mathcal{H} = \mathbb{C}^2.$$

Dirac (Bra–Ket) Notation

Dirac notation provides a compact mathematical language for quantum mechanics.

Ket (Column Vector)

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

Bra (Row Vector)

$$\langle\psi| = [\alpha^* \quad \beta^*]$$

where * denotes the complex conjugate.

$$\boxed{\langle\psi| = (|\psi\rangle)^\dagger}$$

Inner Product and Orthonormality

The inner product between two quantum states is

$$\langle \phi | \psi \rangle.$$

For computational basis states,

$$\langle 0|0\rangle = 1, \quad \langle 1|1\rangle = 1,$$

$$\langle 0|1\rangle = 0, \quad \langle 1|0\rangle = 0.$$

Therefore,

- Basis states are normalized.
- Different basis states are orthogonal.
- Together they form an orthonormal basis.

State Space of Multiple Qubits

Each additional qubit doubles the dimension of the Hilbert space.

Number of Qubits	Hilbert Space
1	$\mathbb{C}^2$
2	$\mathbb{C}^4$
3	$\mathbb{C}^8$
n	$\mathbb{C}^{2^n}$

This exponential growth gives quantum computers their enormous computational potential.

Multi-Qubit Basis States

For two qubits, the computational basis consists of four states.

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

Similarly,

Three qubits produce

$$8$$

basis states,

while

$$n$$

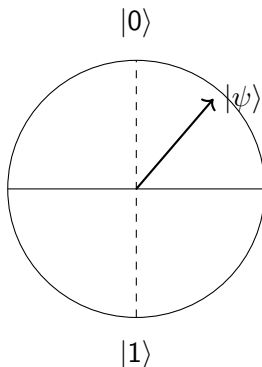
qubits produce

$$2^n$$

basis states

Geometry of Quantum States

A normalized single-qubit state can be visualized on the Bloch Sphere.



The Bloch sphere provides a geometric interpretation of all possible pure single-qubit states.

Worked Example

Given

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle,$$

verify whether the state is normalized.

$$\langle\psi|\psi\rangle = \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{3}{4} + \frac{1}{4} = 1.$$

Hence,

$$|\psi\rangle$$

is a valid quantum state.

Summary and Concept Check

Summary

- Quantum states are represented using Dirac notation.
- Bras are conjugate transposes of kets.
- Inner products determine orthogonality.
- Computational basis states form an orthonormal basis.
- The Hilbert-space dimension grows as 2^n .
- Multi-qubit systems occupy exponentially larger state spaces.

Concept Check

- 1 What is the difference between a bra and a ket?
- 2 Why are $|0\rangle$ and $|1\rangle$ orthonormal?
- 3 Why does the Hilbert-space dimension become 2^n ?
- 4 Why is normalization essential?

Learning Outcomes

After this lecture, you will be able to

- Explain the purpose of the Bloch sphere.
- Represent a single-qubit state geometrically.
- Identify computational basis states.
- Interpret the angles θ and ϕ .
- Relate quantum gates to rotations on the Bloch sphere.

Why Do We Need the Bloch Sphere?

A general single-qubit state is written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where

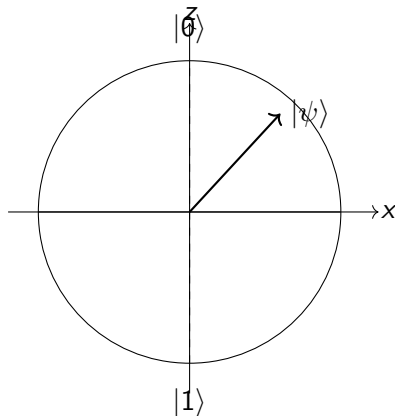
$$|\alpha|^2 + |\beta|^2 = 1.$$

Although mathematically correct, complex vectors are difficult to visualize.

Key Idea

The Bloch sphere provides a geometric representation of every normalized pure single-qubit state.

Structure of the Bloch Sphere



Every point on the surface of the Bloch sphere represents one normalized single-qubit state.

General Representation of a Qubit

Any pure qubit state can be expressed as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle.$$

where

- θ = polar angle
- ϕ = azimuthal angle

The pair (θ, ϕ) uniquely specifies the position of a qubit on the Bloch sphere.

Important Quantum States on the Bloch Sphere

Quantum State	Location
$ 0\rangle$	North Pole
$ 1\rangle$	South Pole
$ +\rangle = \frac{ 0\rangle + 1\rangle}{\sqrt{2}}$	Positive x-axis
$ -\rangle = \frac{ 0\rangle - 1\rangle}{\sqrt{2}}$	Negative x-axis
$ +i\rangle = \frac{ 0\rangle + i 1\rangle}{\sqrt{2}}$	Positive y-axis
$ -i\rangle = \frac{ 0\rangle - i 1\rangle}{\sqrt{2}}$	Negative y-axis

Quantum Gates as Rotations

Quantum gates rotate the state vector on the Bloch sphere.

Pauli-X	Rotation about the x -axis
Pauli-Y	Rotation about the y -axis
Pauli-Z	Rotation about the z -axis
Hadamard	Maps the z -basis to the x -basis

Interpretation

Quantum computation can be viewed as a sequence of rotations of the state vector.

Worked Example

Given

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

identify its position on the Bloch sphere.

Solution:

$$\theta = \frac{\pi}{2}, \quad \phi = 0.$$

Therefore,
the state lies on the positive x -axis and is denoted by

$$|+\rangle.$$

Applications

- Visualizing quantum states
- Understanding quantum gates
- Quantum state tomography
- Quantum communication
- Quantum error correction

Limitation

The Bloch sphere represents only a single pure qubit.
Multi-qubit systems require higher-dimensional Hilbert spaces.

Summary and Concept Check

Summary

- The Bloch sphere gives a geometric picture of a qubit.
- Every pure qubit corresponds to one point on the sphere.
- θ and ϕ determine the quantum state.
- Quantum gates correspond to rotations.
- The Bloch sphere is applicable only to single-qubit pure states.

Concept Check

- 1 Why is the Bloch sphere useful?
- 2 Where are $|0\rangle$ and $|1\rangle$ located?
- 3 What do θ and ϕ represent?
- 4 Why can the Bloch sphere not represent two-qubit states?

Learning Outcomes

After this lecture, you will be able to

- Explain the No-Cloning Theorem.
- Compare copying of classical and quantum information.
- Understand why unknown quantum states cannot be copied.
- Follow a simple proof of the theorem.
- Explain the role of the theorem in quantum cryptography.

Classical Copying vs Quantum Copying

Classical Information

$$0 \rightarrow 00$$

$$1 \rightarrow 11$$

Classical bits can be copied perfectly any number of times.

Quantum Information

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Can we produce

$$|\psi\rangle|\psi\rangle ?$$

This question leads to the No-Cloning Theorem.

Statement of the No-Cloning Theorem

No-Cloning Theorem

It is impossible to construct a universal machine that can produce an identical copy of an arbitrary unknown quantum state.

Important observations

- The theorem applies to **unknown** quantum states.
- It applies to **all arbitrary** quantum states.
- A known state can be prepared again, but it cannot be cloned by a universal copying machine.

How Would a Quantum Copier Work?

Assume a perfect quantum copying machine exists.

Initially,

$$|\psi\rangle|0\rangle$$

where the second qubit is blank.

After cloning,

$$|\psi\rangle|0\rangle \longrightarrow |\psi\rangle|\psi\rangle$$

Similarly,

$$|\phi\rangle|0\rangle \longrightarrow |\phi\rangle|\phi\rangle$$

Can the same machine copy every possible quantum state?

Proof of the No-Cloning Theorem

Assume a unitary operator U performs cloning.

$$U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle$$

$$U(|\phi\rangle|0\rangle) = |\phi\rangle|\phi\rangle$$

Taking inner products,

$$\langle\psi|\phi\rangle = (\langle\psi|\phi\rangle)^2$$

This equation is true only when

$$\langle\psi|\phi\rangle = 0$$

or

$$\langle\psi|\phi\rangle = 1.$$

Therefore,

Physical Interpretation

Why is cloning impossible?

- A quantum state cannot be determined exactly without measurement.
- Measurement generally disturbs the quantum state.
- A cloning machine would need complete knowledge of the state without changing it.
- This contradicts the principles of quantum mechanics.

Key Idea

Quantum information cannot be copied without fundamentally affecting the system.

Applications of the No-Cloning Theorem

The No-Cloning Theorem is fundamental to modern quantum technologies.

- Quantum Key Distribution (BB84)
- Quantum Cryptography
- Secure Quantum Communication
- Quantum Networks
- Quantum Repeaters

Because qubits cannot be copied perfectly, an eavesdropper cannot secretly duplicate transmitted quantum information.

Worked Example

Alice sends the state

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Eve intercepts the qubit and attempts to create an identical copy.

Can Eve produce

$$|\psi\rangle|\psi\rangle$$

without disturbing the original state?

Answer: No.

Any attempt to obtain complete information about an unknown quantum state will disturb the state and reveal the presence of an eavesdropper.

Summary and Concept Check

Summary

- Classical information can be copied.
- Unknown quantum states cannot be cloned.
- No universal quantum copying machine exists.
- The proof follows from the linearity of quantum mechanics.
- The theorem is the foundation of quantum cryptography.

Concept Check

- 1 Why can classical bits be copied but qubits cannot?
- 2 Why does the theorem apply only to unknown quantum states?
- 3 How does the No-Cloning Theorem improve communication security?
- 4 Which quantum technology relies directly on this theorem?

Learning Outcomes

After this lecture, you will be able to

- Explain why density matrices are required.
- Distinguish between pure and mixed quantum states.
- Construct a density matrix from a state vector.
- Identify the important properties of density matrices.
- Describe applications of density matrices in quantum computing.

Why Do We Need Density Matrices?

A pure quantum state is represented by

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

However, real quantum systems often

- interact with the environment,
- lose coherence,
- exist as statistical mixtures.

Key Idea

Density matrices provide a unified description of both **pure** and **mixed** quantum states.

Pure States vs Mixed States

Pure State

Complete knowledge of the system

Represented by a state vector

Contains quantum coherence

Example: $|+\rangle$

Mixed State

Statistical uncertainty

Represented by a density matrix

Represents a probability mixture

Example:

50% $|0\rangle$

50% $|1\rangle$

Note:

A mixed state is **not** the same as a quantum superposition.

Definition of a Density Matrix

For a pure quantum state,

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

its density matrix is defined as

$$\rho = |\psi\rangle\langle\psi|$$

The density matrix is obtained by taking the **outer product** of the state vector with its conjugate transpose.

For a single qubit,

$$\rho = \begin{bmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{bmatrix}.$$

Example: Density Matrix of a Pure State

Consider the quantum state

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Column vector

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Density matrix

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

The off-diagonal elements indicate quantum coherence.

Example: Density Matrix of a Mixed State

Suppose a source prepares

- $|0\rangle$ with probability 0.5
- $|1\rangle$ with probability 0.5

Then,

$$\rho = 0.5|0\rangle\langle 0| + 0.5|1\rangle\langle 1|$$

which becomes

$$\rho = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Notice that the off-diagonal elements are zero.

Properties of Density Matrices

Every valid density matrix satisfies the following properties.

- 1 Hermitian

$$\rho^\dagger = \rho$$

- 2 Unit Trace

$$\text{Tr}(\rho) = 1$$

- 3 Positive Semi-definite

$$\lambda_i \geq 0$$

- 4 Eigenvalues satisfy

$$0 \leq \lambda_i \leq 1.$$

These conditions ensure that ρ represents a physical quantum state.

Applications of Density Matrices

Density matrices are indispensable in modern quantum technologies.

- Open quantum systems
- Quantum decoherence
- Quantum communication
- Quantum error correction
- Noisy quantum computers
- Quantum state tomography

Key Message

Density matrices allow us to describe realistic quantum systems, not just ideal isolated qubits.

Worked Example

Find the density matrix corresponding to

$$|\psi\rangle = |0\rangle.$$

Solution

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\rho = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Check:

$$\text{Tr}(\rho) = 1.$$

Hence, this is a valid pure quantum state.

Summary and Concept Check

Summary

- Density matrices generalize the state-vector formalism.
- Pure states satisfy

$$\rho = |\psi\rangle\langle\psi|.$$

- Mixed states represent statistical ensembles.
- Off-diagonal elements describe quantum coherence.
- Density matrices are widely used in quantum computing and quantum communication.

Concept Check

- 1 Why are density matrices needed?
- 2 What is the difference between a pure state and a mixed state?
- 3 What do the off-diagonal elements represent?
- 4 Why must the trace of a density matrix be equal to 1?

Learning Outcomes

After this lecture, you will be able to

- Define a mixed quantum state.
- Distinguish between pure states and mixed states.
- Construct the density matrix of a mixed state.
- Explain the physical origin of mixed states.
- Recognize the role of mixed states in quantum computing.

What is a Mixed State?

In an ideal quantum computer, qubits are prepared in well-defined **pure states**.

In practice, quantum systems interact with their surroundings, introducing uncertainty.

A **mixed state** represents a statistical ensemble of different quantum states with known probabilities.

Definition

A mixed state describes incomplete knowledge about the exact quantum state of a system.

Pure States vs Mixed States

Pure State

Complete information

Single state vector

Quantum superposition possible

Maximum coherence

Ideal isolated system

Mixed State

Statistical uncertainty

Probability distribution of states

No coherent superposition

Reduced or no coherence

Realistic noisy system

Important:

A mixed state is a probabilistic mixture, not a quantum superposition.

Mathematical Representation

Suppose the system is

- in state $|\psi_1\rangle$ with probability p_1
- in state $|\psi_2\rangle$ with probability p_2
- ...

Then the density matrix is

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$$

where

$$\sum_i p_i = 1.$$

This is the general representation of a mixed quantum state.

Example of a Mixed State

Suppose

$$P(|0\rangle) = 0.7,$$

$$P(|1\rangle) = 0.3.$$

Then

$$\rho = 0.7|0\rangle\langle 0| + 0.3|1\rangle\langle 1|.$$

Hence,

$$\rho = \begin{bmatrix} 0.7 & 0 \\ 0 & 0.3 \end{bmatrix}.$$

Notice that all off-diagonal elements are zero.

Mixed State vs Quantum Superposition

Consider

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Its density matrix is

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Now compare with the mixed state

$$\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The off-diagonal elements represent **quantum coherence**.

How Do Mixed States Arise?

Mixed states naturally occur because of

- Interaction with the environment
- Measurement uncertainty
- Noise in quantum devices
- Loss of coherence (decoherence)
- Imperfect state preparation

Modern quantum computers spend significant effort minimizing these effects.

Applications of Mixed States

Mixed-state formalism is essential for

- Quantum communication
- Quantum cryptography
- Quantum error correction
- Noisy Intermediate-Scale Quantum (NISQ) devices
- Quantum sensing
- Open quantum systems

Mixed states allow realistic modeling of quantum hardware.

Worked Example

A qubit is

- in $|0\rangle$ with probability 0.6
- in $|1\rangle$ with probability 0.4

Find the density matrix.

Solution

$$\rho = 0.6|0\rangle\langle 0| + 0.4|1\rangle\langle 1| = \begin{bmatrix} 0.6 & 0 \\ 0 & 0.4 \end{bmatrix}.$$

Check:

$$\text{Tr}(\rho) = 1.$$

Summary and Concept Check

Summary

- Mixed states represent statistical uncertainty.
- They are described using density matrices.
- Mixed states differ fundamentally from superposition.
- Off-diagonal elements indicate quantum coherence.
- Mixed states model realistic quantum systems.

Concept Check

- 1 What is the difference between a pure state and a mixed state?
- 2 Why are mixed states represented by density matrices?
- 3 What do zero off-diagonal elements indicate?
- 4 Why are mixed states important in quantum computing?

Tutorial Objectives

After completing this tutorial, you should be able to

- Construct density matrices for pure states.
- Construct density matrices for mixed states.
- Distinguish between pure and mixed states.
- Verify basic properties of density matrices.
- Solve introductory problems independently.

Quick Review

Recall the key equations.

Pure state:

$$\rho = |\psi\rangle\langle\psi|$$

Mixed state:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

Properties of a valid density matrix:

- Hermitian
- Trace equals 1
- Positive semi-definite

Worked Example 1

Construct the density matrix for

$$|\psi\rangle = |0\rangle.$$

Solution

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\rho = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Verify

$$\text{Tr}(\rho) = 1.$$

Worked Example 2

Find the density matrix of

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Solution

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\rho = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Notice the non-zero off-diagonal elements, indicating quantum coherence.

Worked Example 3

A source produces

- $|0\rangle$ with probability 0.7
- $|1\rangle$ with probability 0.3

Find the density matrix.

Solution

$$\rho = 0.7|0\rangle\langle 0| + 0.3|1\rangle\langle 1| = \begin{bmatrix} 0.7 & 0 \\ 0 & 0.3 \end{bmatrix}.$$

This is a mixed state.

Practice Problems

Solve the following.

Problem 1

Find the density matrix of

$$|1\rangle.$$

Problem 2

Find the density matrix of

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Problem 3

A qubit is

80% in $|0\rangle$

20% in $|1\rangle$

Construct the density matrix

Conceptual Questions

Answer the following.

- 1 Why are density matrices required?
- 2 What is the difference between a pure state and a mixed state?
- 3 What do the diagonal elements represent?
- 4 What do the off-diagonal elements represent?
- 5 Why must the trace of a density matrix always be equal to one?

Challenge Problems

Attempt these independently.

- 1 Show that

$$\rho = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

represents a pure state.

- 2 Explain why

$$\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

represents a mixed state.

- 3 Compare both matrices and identify the role of coherence.

Summary and Take-Home Exercise

Summary

- Density matrices describe both pure and mixed states.
- Pure states are obtained using $\rho = |\psi\rangle\langle\psi|$.
- Mixed states are weighted averages of pure-state density matrices.
- Off-diagonal elements represent quantum coherence.

Take-Home Exercise

Construct the density matrices for

$$|+\rangle, |-\rangle, |+\hat{i}\rangle, |-\hat{i}\rangle$$

and determine whether each represents a pure or mixed state.